A Sequential Stochastic Approach to Rationalizing Hedging Decisions in Agricultural Markets

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May 11, 2026

Abstract

Farmers have historically hedged much less of their corn production than economists recommended. Previous research has mostly modeled farmer behavior by employing the mean-variance hedge ratio in a 2-period investment setting. In this paper, we eschew this approach and use expected utility theory to rationalize hedging decisions made by farmers in a dynamic setting. We enable our farmer to make hedging decisions on a weekly basis to account for incremental changes in prior market beliefs. We operationalize farmer uncertainty through both Gaussian and non-Gaussian distributions. We find that subjective prior beliefs - especially prior mean and risk aversion, play a significant role in hedging decisions for all distributions tested. Other factors such as initial wealth play little to no role in optimal hedging decisions. Our results indicate that the heterogeneous hedging behavior of farmers can be fully rationalized through our expected utility framework.

Keywords: hedging; risk; CRRA; futures.

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1 Introduction

Hedging behavior in farmers has been seen as low compared to what conventional financial wisdom would suggest. Data from Prager et al. (2020) show that, in 2016, just 9.9% of corn and soybean producing farms in the United States hedged their corn product. The farmers that hedged their corn sold on average 41.1% of their product on the futures market. Only 6% of farmers invested in the futures market hedged the entirety of their crop. Moreover, larger corn farms are much more likely to trade in options.

Dorfman and Karali (2010) find that habit plays a significant role in hedging decisions, defined as the connection between current and lagged hedging decisions. High basis risk and greater cash constraints are also connected with lower levels of hedging (Prager et al. (2020)). Farmers with more certainty about prices, lower levels of risk aversion and diversified investment portfolios tend to hedge less (Pannell et al. (2008)). Larger farms tend to trade more frequently in futures and options, along with younger farmers and farmers with college educations (MacDonald (2020)). Pennings and Leuthold (2000) found that risk attitude played a significant role in determining the extent to which farmers hedged in the futures market.

Previous literature on hedging decisions focuses significantly on the mean-variance hedge ratio framework, namely:

$$h^* = \frac{Cov(\Delta S, \Delta F)}{Var(\Delta F)} \tag{1}$$

where ΔS = the change in spot price and ΔF = the change in futures prices. This methodology was proposed by Johnson (1960), positing that the solution is the optimal hedge ratio. The popularity of this model accelerated as it garnered empirical validity (Ederington (1979)).

However, this deterministic formula fails to account for subjective measures such as risk preference and prior beliefs regarding the direction and magnitude of future price changes throughout the season. In this paper, we attempt to further rationalize hedging behavior in a sequential dynamic stochastic setting that accounts for subjective beliefs about the projected futures prices throughout the season. We endow our representative farmer with a constant relative risk aversion (CRRA) expected utility function that is optimized weekly over 27 weeks, which is the hedging window of the December corn contract.

Subjective probability can be dated back to Ramsey's *Truth and Probability* (Ramsey (1931)), where he proposed that probability is not objective in the frequentist sense, but expressed as a subjective degree of belief contingent upon prior beliefs. In *General Theory* (Keynes (1936)), Keynes argued that the "animal spirits", or one's subjective economic outlook, played a heavy role in one's investment decisions. Many previous hedging models fail to account for said subjectivity. Implementing subjective parameters into our expected utility model will be key in rationalizing these investment decisions.

Many previous papers model structural hedging decisions in a two-period time setting (Lapan and Moschini (1994), Pannell et al. (2008)). Hedging was later introduced through a dynamic sequential framework, allowing for the characteization of a time path towards an optimal hedge (Anderson and Danthine (1983)). Our sequential dynamic approach introduces the assumption that farmers' hedging habits might change on a week by week basis as their prior beliefs on the state of the corn market change. This relaxes the restrictive assumptions of the two period model and enables farmers to adjust hedging decisions as new information arrives throughout the growing season. Myers and Hanson (1996) extended Anderson and Danthine (1983)'s analysis by introducing a discrete dynamic hedging problem that relaxes assumptions regarding the underlying utility function and price distributions.

In our structural model, the farmer gains utility from a certain futures price, uncertain futures price later on in the growing season, previous futures sales, the expectation of the spot price at the conclusion of the season, and initial wealth. The flexible nature of the CRRA utility function enables us to fluctuate the Arrow-Pratt coefficient of relative risk aversion to assess how risk aversion levels affect aggregate hedging decisions over a futures contract season. At t = 1, the farmer will optimize his utility over the current week and the remaining hedging season, where each week from 2 to 27 is denoted with a distribution to represent the farmer's beliefs about the future prices. At t = 2, the farmer's week 1 decision will be internalized as he plans from week 3 to 27. This process iterates until week 27 when the contract expires, or the farmer has hedged the entirety of their corn. Additionally, we operate under a non-stochastic production function to isolate the effects of risk aversion and other structural parameters of interest.

Previous papers such as Lapan and Moschini (1994) assume normal distributions of futures prices predictions for analytical convenience; however, this fails to take into account the impact of higher moments. For instance, Gordon (1985) ran a variety of normality tests on the price distributions in futures markets of many different agricultural commodities and rejected normality in every instance. Baillie and Myers (1991) modeled price changes in corn futures markets within the GARCH framework and found the price changes to be non-normal.

In light of this literature, it is imperative to jettison the normality assumption of price distributions for futures markets. Using Fleishman's power method (Fleishman (1978)), we will employ both Gaussian and non-Gaussian distributions as subjective priors. Fleishman's power method offers the flexibility of moment-preserving spreads of a normal distribution. Namely, within sensible bounds, mean, variance, skewness and kurtosis can all be fluctuated to desired values while matching other moments. These transformations can help us isolate the effects of each moment on the farmer's hedging decisions, *ceteris paribus*.

For Gaussian distributions, we will fluctuate both mean and standard deviation of our representative farmer's distribution of future expected corn prices to assess their impact on final hedging decisions. Mean represents a farmer's level of bearishness or bullishness – subtracting from the mean denotes bearishness, whereas adding to the mean represents bullishness. Similarly, standard deviation represents subjective uncertainty. We also account for non-Gaussian priors as well. This enables us to employ priors that are both skewed and kurtotic in structure. Using Fleishman's power method, we create a variety of different non-

Gaussian priors that are both mean and variance preserving to isolate the effects of these higher moments on hedging decisions.

To our knowledge, this is the first paper that rationalizes farmers' hedging decisions in a dynamic framework that allows for subjective parameters in the agent's expected utility function. Moreover, we isolate the effects of changes in moments of the agent's belief distribution and report their effects on the optimal hedge ratio at the end of the season. This allows us to model hedging decisions in a more accurate way to rationalize farmers' hedging decisions.

2 Modeling Price Expectations

In our model, we use the December 2024 corn futures contract (ZCZ24) from the Chicago Board of Exchange. While futures prices are often expressed in cents, we denote the futures price in dollars for computational purposes. Figure 1 shows a boxplot of the weekly 2024 corn futures prices in dollars.

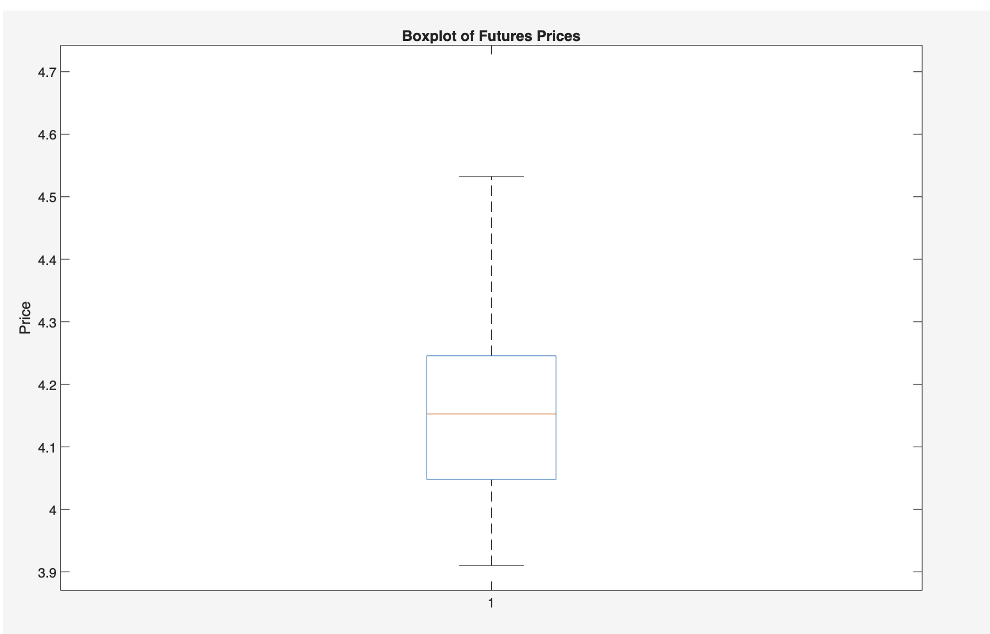

Figure 1: Boxplot of 2024 Futures Prices, December Contract

For our model, $t = 1$ is the last week of May, and $T = 27$ is the end of the December contract. These prices serve as the certain payment received in the farmer's utility function. Moreover, they serve as a mean for our prior distributions. We employ the Fleishman Power Method to generate non-Gaussian prior distributions that satisfy moment conditions. The Fleishman Power method is a way to generate non-Gaussian distributions with given moments by transforming a standard normal distribution with a system of polynomial equations. Specifically, let γ_1 be the desired level of skewness and γ_2 be the desired level of (excess) kurtosis [1]. Given γ_1 and γ_2, solving the system of equations consisting of F_1, F_2, F_3 provides us with coefficients a, b, c, d which are necessary to facilitate the Fleishman power method. The system of equations is defined as:

$$F_1(b, c, d) = b^2 + 6bd + 2c^2 + 15d^2 - 1 = 0 \tag{2}$$

$$F_2(b, c, d) = 2c(b^2 + 24bd + 105d^2 + 2) - \gamma_1 = 0 \tag{3}$$

$$F_3(a, b, c) = 24(bd + c^2[1 + b^2 + 28bd] + d^2[12 + 48bd + 141c^2 + 225d^2]) - \gamma_2 = 0 \tag{4}$$

Let $X \sim N(0, 1)$. After solving the system of equations for the unknown paramaters a, b, c, d with given γ_1, γ_2, coefficients a, b, c, d are plugged into the cubic equation to give us our distribution.

$$\tilde{Y} = a + bX + cX^2 + dX^3 \tag{5}$$

Given $\tilde{Y}$ from (5), adding the desired mean, μ, and multiplying the desired variance, σ, gives us the final moment-preserving distribution [2], defined as

$$Y = \mu + \tilde{Y} * \sigma \tag{6}$$

[1] All mentions of kurtosis in this paper will refer to excess kurtosis for brevity. Normal distributions have a kurtosis level of 3, but an excess kurtosis of 0.

[2] This generation process is also amenable to Gaussian distributions. When γ_1 and $\gamma_2 = 0$, a = 0, b = 1, c = 0, d = 0 and (6) changes to $Y = \mu + X\sigma$.

For any given week t, we assume the mean of each forecasted distribution is the current futures price f_t. For the forecasted prior variance, we use historical standard deviations of futures price differences for the period of 2000 to 2020 in the December contract. For example, for week 2's prior in t = 1, the standard deviation will be the historical standard deviation between the price differences of the futures prices in week 1 and week 2. The distributions of the expected futures prices need not be correlated with each other, as Samuelson (2015) proved that the next period's futures price differences are uncorrelated with the previous period's prices. Let t equal the current time period, and j equal a forward looking time horizon for weeks later on in the season. Namely,

$$\sigma_{t,j} = \sqrt{\frac{1}{n-1} \sum_{y=1}^{n} \left[\left(f_{t+j}^{(y)} - f_t^{(y)} \right) - \overline{(f_{t+j} - f_t)} \right]^2} \tag{7}$$

This ensures that the farmer's prior variance starts out small in short time horizons, and expands as the time horizon extends towards T. Indeed, a rational farmer will have more certainty about next week's futures price compared to futures prices later on in the season.

In our model, the representative farmer also has the option to sell their corn in the spot market when the December corn futures contract expires. Determining the distribution for the spot price of corn is a non-trivial task. Unlike futures prices, there is no single spot price for corn in the United States. Given that the spot price exists after the December futures contract has expired, we need to approximate it. Fama and French (1987) appeal to the Efficient Market Hypothesis and show that the current futures price is a strong, unbiased predictor of the spot price for any given commodity. Gardner (1976) extended this work to agricultural commodities as well. In light of these findings, we use the current futures price at time t as the mean for the spot price distribution. To determine the variance of the distribution of the spot price, we forecast out the historical variance of futures price differences in (7) using a linear regression model. For example, when t = 1, we have 26 forecasted futures price variances courtesy of (7). From there, we run a bivariate linear regression with t on the x-axis and σ on the y-axis. Figure 2 illustrates what this looks like when $t = 1$. After obtaining the linear regression coefficient, we forecast out the spot price variance to the terminal time period. Because the variance of (7) is increasing from t = 1 to t = 26, this ensures that the spot price has a high degree of variance.

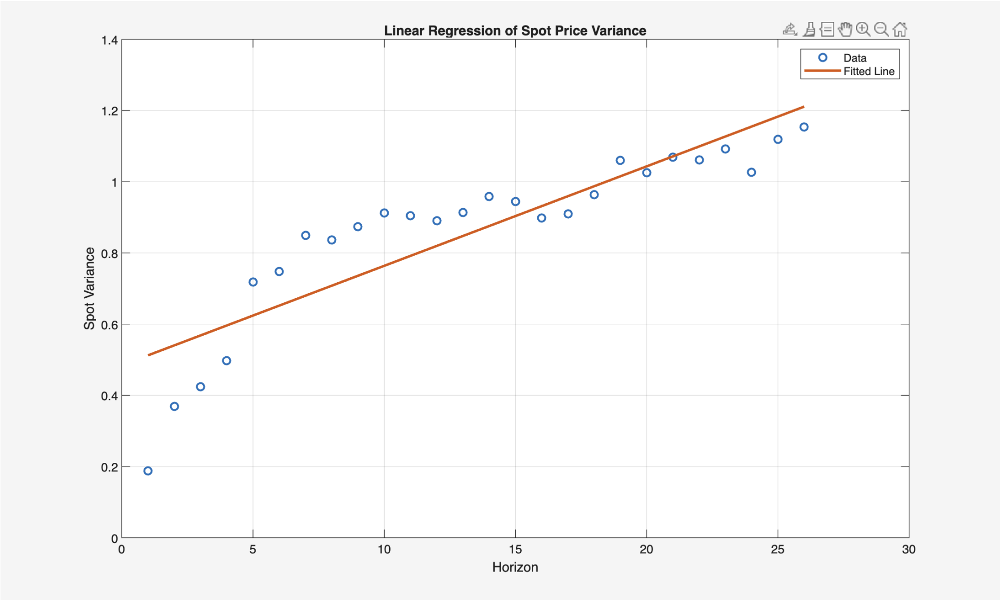

Figure 2: Forecasting spot price variance at $t = 1$

Let $\tilde{f}_j$ be the farmer's prior for futures prices and $\tilde{P}_j$ be the prior distribution for the spot price at the end of the season. The farmer has a prior distribution for each forecasted week in the season. Thus, at time period t, the farmer has a total of $T - t$ prior distributions. At time t, the current futures price is f_t. The standard deviation of the distribution at a later time period j, is given as $\sigma_{t,j}$ from equation (7) above, and the standard deviation for the spot price, σ_{OLS}, is specified above in Figure 2. γ_1 and γ_2 are skewness and kurtosis, respectively. We also address the fact that not all farmers have the same level of uncertainty. To do so, we introduce an uncertainty parameter, κ, that multiplies the variance to express different degrees of uncertainty. Lower levels of κ imply high levels of certainty about the trajectory of futures prices, whereas higher levels of κ represent uncertainty. For a given future time period j, the farmer's prior distribution assumes the form:

$$\tilde{f}_j \sim Fleishman(f_t, \kappa * \sigma_{t,j}, \gamma_1, \gamma_2) \tag{8}$$

$$\tilde{P}_j \sim Fleishman(f_t, \kappa * \sigma_{OLS}, \gamma_1, \gamma_2) \tag{9}$$

Figures 3 and 4 illustrate the powerful moment preserving effects of the Fleishman power method. For both figures, we set the mean, μ, to equal 4.20, which is the mean of the corn futures prices in 2024. Figure 3 shows how skewness can be fluctuated while preserving other moments. In Figure 3, The blue, left skewed distribution has the following prespecified moments: $\mu = 4.20, \sigma = 1, \gamma_1 = -0.8$, and $\gamma_2 = 0$. The orange, right skewed distribution has the moments: $\mu = 4.20, \sigma = 1, \gamma_1 = 0.8$, and $\gamma_2 = 0$. Figure 4 illustrates how kurtosis can be varied while holding other moments constant. The orange distribution has the following prespecified moments: $\mu = 4.20, \sigma = 1, \gamma_1 = 0$, and $\gamma_2 = -1$, and the blue distribution has the moments of: $\mu = 4.20, \sigma = 1, \gamma_1 = 0$, and $\gamma_2 = 1$.

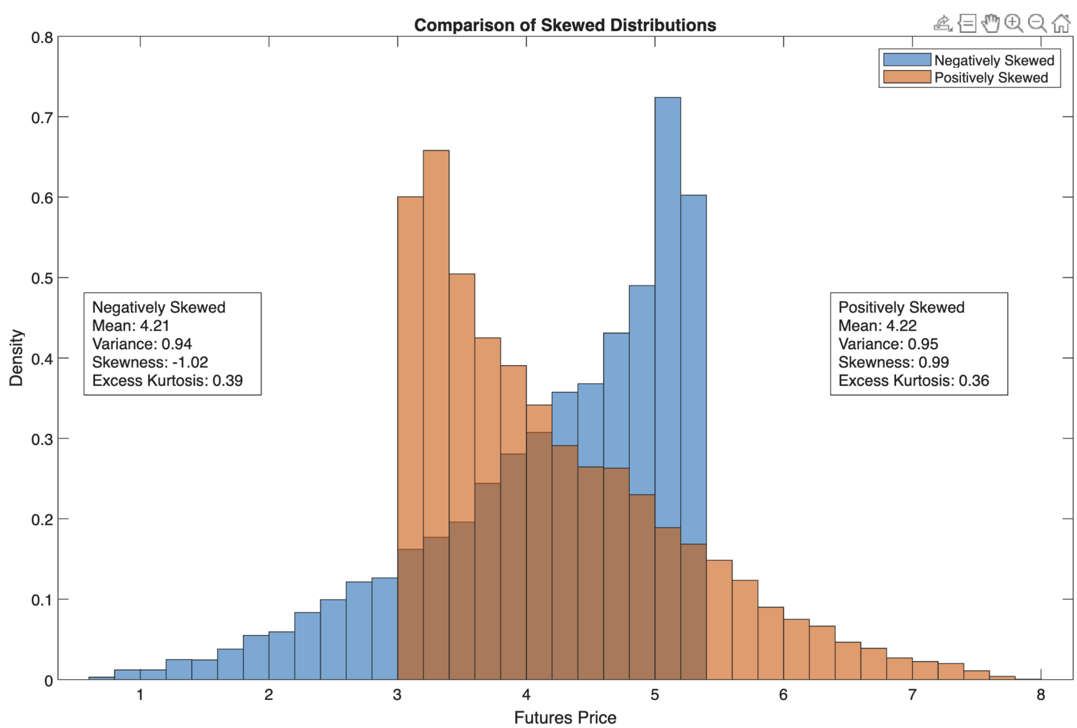

Figure 3: Moment Preserving Skew Adjustment

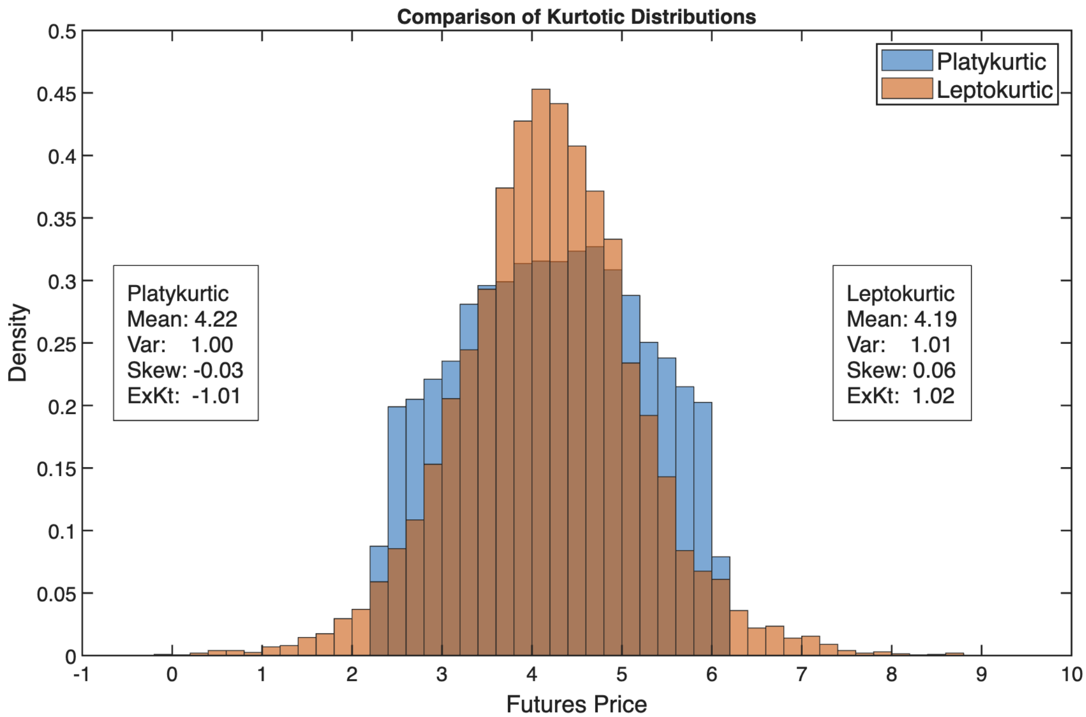

Figure 4: Moment Preserving Kurtosis Adjustment

3 Model

Let f_t denote the futures price at week t. At each t, the farmer has two choice variables, θ_t and $\{\tilde{\theta}_j\}_{j=t+1}^T$, where θ_t represents the quantity of the crop to be hedged in week t, and $\tilde{\theta}_j$ represents the quantity the farmer plans to hedge in later periods. Both θ_t and $\{\tilde{\theta}_j\}_{j=t+1}^T$ are in percentage terms. The $\tilde{\theta}_j$ parameters are not locked into the farmer's final hedging decision, only θ_t is. However, $\tilde{\theta}_j$ helps the farmer plan and internalize future hedging decisions. For instance, in $t = 1$, if a farmer decides to hedge 10% of their corn in the current period (θ_t), then the farmer will be able to hedge 90% of their corn in later time periods, and 0.90 will be the new upper bound of the constraint moving forward.

The parameter o expresses the subjective degree of confidence that the farmer has in the future trajectory of the futures price. The parameter o increases on a linear basis throughout the iterations. Specifically, if $o = 0.01$, then 1 cent is added to each additional week's forecasted mean, so if $o = 0.01$, t = 2 will have 1 cent added to the forecasted mean, t = 3 will have 2 cents added to the forecasted mean, and so forth, culminating with 27 cents being added to the predicted spot price. This represents the bullishness of the subjective probability. The distribution $\tilde{f}_j$ is generated from the Fleishman power method detailed in section 2. Each $\tilde{f}_j$ distribution contains 10,000 draws of predicted prices. W_0 represents the initial wealth of our representative farmer. The farmer gains utility from current futures purchases, later planned purchases, and previous sales on the futures market. The farmer also has the option to forego hedging and sell at the final period on the spot market. We denote this predicted terminal spot price as $\tilde{P}_T$. Each $\tilde{P}_T$ distribution contains 10,000 draws of predicted prices generated from the Fleishman power method. We will examine all four moments of $\tilde{f}_j$ and $\tilde{P}_j$ to assess their effect on hedging decisions for farmers. Throughout the season, the farmer solves 27 optimization problems, one for each week. The hedge ratio at the end of the season, $T = 27$ is defined as $\sum_{t=1}^{T} \theta_t$.

Our representative farmer is endowed with a CRRA utility function, which he maximizes by choosing weekly hedging positions according to: [1]

$$
\begin{aligned}
\max_{\theta_t \, \{\tilde{\theta}_j\}_{j=t+1}^{T}} \quad & \mathbb{E}_t \left[\frac{1}{1-\rho} \left(W_0 + \underbrace{f_t \theta_t}_{\substack{\text{Current} \\ \text{futures price}}} + \underbrace{\sum_{k=1}^{t-1} f_k \theta_k}_{\substack{\text{Previous} \\ \text{sales}}} + \underbrace{\sum_{j=t+1}^{T-1} (\tilde{f}_j + o)\tilde{\theta}_j}_{\substack{\text{Predicted} \\ \text{prices}}} + \underbrace{\left(1 - \sum_{k=1}^{t} \theta_k - \sum_{j=t+1}^{T-1} \tilde{\theta}_j \right) (\tilde{P}_T + o)}_{\substack{\text{Spot} \\ \text{price}}} \right)^{1-\rho} - 1 \right] \\
\text{subject to} \quad & \sum_{k=1}^{t} \theta_k + \sum_{j=t+1}^{T-1} \tilde{\theta}_j \le 1, \\
& \theta_t \ge 0, \quad \tilde{\theta}_j \ge 0.
\end{aligned}
\tag{10}
$$

[1]Note that $E[U]$ decomposes to $E[log(U)]$ when $\rho = 1$, by L'Hôpital's Rule.

Let R be the revenue flow nested in the CRRA utility function. After establishing the Lagrangian, the first order conditions for optimal maxima are:

$$\frac{\partial \mathcal{L}}{\partial \theta_t} = \mathbb{E}[(f_t - \tilde{P}_T)\, R^{-\rho}] = 0 \tag{11}$$

$$\frac{\partial \mathcal{L}}{\partial \tilde{\theta}_j} = \mathbb{E}[[(\tilde{f}_j + o) - \tilde{P}_T]\, R^{-\rho}] = 0 \tag{12}$$

4 Comparative Statics

Our farmer gains utility from current wealth, a sale in the futures market at price f_t, previous sales, and futures prices that are subjectively projected later on in the season $\tilde{f}_j$ Because f_t is strictly positive, the farmers' utility is strictly increasing in f_t. However, because $\tilde{f}_j$ is a random variable, we can apply a 4th order Taylor series around the mean, $\mu = f_t$, to approximate $E[U(\tilde{f}_j)]$ as a summation of its moments. Specifically, if we let $\mu = f_t$

$$E[U(\tilde{f_j})] \approx E[U(f_t) + U'(f_j)(\tilde{f}_j - f_t) + \frac{1}{2}U''(f_t)(\tilde{f}_j - f_t)^2 + \frac{1}{6}U'''(f_t)(\tilde{f}_j - f_t)^3 + \frac{1}{24}U^{(4)}(f_t)(\tilde{f}_j - f_t)^4] \tag{13}$$

$$\approx U(f_t) + U'(f_t)\,\underbrace{E[\tilde{f}_j - f_t]}_{=0} + \frac{1}{2}U''(f_t)\,\underbrace{E[\tilde{f}_j - f_t]^2}_{=\sigma^2} + \frac{1}{6}U'''(f_t)\,\underbrace{E[\tilde{f}_j - f_t]^3}_{=\gamma_1} + \frac{1}{24}U^{(4)}(f_t)\,\underbrace{E[\tilde{f}_j - f_t]^4}_{=\gamma_2} \tag{14}$$

Next, we analyze the effect of each higher moment by taking each higher order derivative and plugging them into (14).

4.1 Risk Aversion

Because the first expectation term washes out, the first order Taylor expansion can be expressed through Jensen's Inequality:

$$E[U(\tilde{f}_j)] \lesseqgtr U(f_t) \tag{15}$$

where the inequality is contingent upon the sign of the ρ parameter. If a utility function is CRRA, then $U''(f_j) < 0$ if $\rho > 0$, and $U''(f_j) > 0$ if $\rho < 0$, In other words, when ρ is positive, the farmer is risk averse, and the utility of the expected value dominates the expected value of the utility. The converse holds for risk loving farmers.

4.2 Variance

From the second order Taylor expansion, variance can be approximated as

$$\frac{1}{2}\mathbb{E}[-\rho R^{-\rho-1}(f_t - \tilde{P}_t)^2] * \sigma^2 \tag{16}$$

For variance, if $\rho > 0$, then $U''(f_t) < 0$. Similarly, if $\rho < 0$, then $U''(f_t) > 0$. Risk averse farmers view prior variance as a threat. Thus, risk averse farmers experience a decrease in utility with increasing variance. As variance increases, we can expect farmers to favor the current, risk free hedge θ_t over the $\tilde{\theta_j}$ distribution with a high degree of variance. As a result, more variance should result in a higher level of hedging. Moreover, ρ enters into (14) exponentially in R as $R^{-\rho-1}$. Thus, we should expect the influence of variance to wane as ρ increases.

4.3 Skewness

From the third order Taylor expansion, skewness can be approximated as

$$\frac{1}{6}\mathbb{E}[\rho(1+\rho)R^{-\rho-2}(f_t-\tilde{P}_t)^3]*\gamma_1 \tag{17}$$

Skewness, a notion of directional bias in a non-Gaussian distribution, has similar results to the change in the mean, f_t. Namely, with the exception of $\rho \in (-1,0)$, utility is increasing with positive skewness. Positive skewness is represented graphically as a right-tailed distribution. Having a longer right-tailed prior is synonymous with having a more positive outlook on later market prices. However, similar to variance, as ρ increases, the utility from the revenue $R^{-\rho-2}$ decreases exponentially, whereas the multiplicative term to the left of the revenue function $\rho(1+\rho)$ increases linearly. Combining the exponential decrease of ρ with the Taylor coefficient of $\frac{1}{6}$, we can expect skewness to have little to no effect on highly risk averse farmers.

4.4 Kurtosis

From the fourth order Taylor expansion, kurtosis can be approximated as

$$\frac{1}{24}\mathbb{E}[-\rho(\rho+1)(\rho+2)R^{-\rho-3}(f_t-\tilde{P}_t)^4]*\gamma_2 \tag{18}$$

Similar to variance, all risk averse agents dislike kurtosis, as $-\rho(\rho+1)(\rho+2) < 0 \quad \forall \quad \rho > 0$. Kurtotic distributions are heavy on the tails, and risk averse agents dislike the potential losses. The Taylor coefficient of $\frac{1}{24}$ combined with the exponential nature of $R^{-\rho-3}$ implies that the farmer likely won't change their optimal hedge ratio very much in the face of kurtosis.

4.5 Initial Wealth

The notion of the CRRA utility function exhibiting Decreasing Absolute Risk Aversion (DARA) is a well-known result in microeconomic theory. Let R be the revenue stream in (10) above and $A(R)$ be the coefficient of absolute risk aversion, defined as $A(R) = -\frac{u''(R)}{u'(R)}$. For CRRA utility, $A(R) = \frac{-\rho}{R}$, and $\frac{\partial A(R)}{\partial R} < 0$, thus satisfying DARA. As a result, we should expect farmers to hedge less when endowed with higher levels of initial wealth.

5 Solving Optimization Model

We optimize equation (10) in MATLAB using the fmincon package with the parameters listed in Table 1[1]. Each parameterization yields 960 distinct combinations. For each of the 960 combinations, 1000 simulations are run and optimized. The average season ending hedge ratio is calculated across all 1000 simulations for each parameter combination.

Table 1 lists the parameters we employ in our optimization model. Initial wealth is calculated using USDA farm statistics. For computational purposes, we exploit the homothetic nature of the CRRA utility function and scale the output to 1 unit and interpret outcomes on a per-unit basis. We choose initial wealth levels of \$25 and \$50 for our representative farmers. We assume that initial wealth is comprised of land valuation, machinery investment and off-farm income, and we utilize USDA farm statistics to calibrate these values. Details of these calculations are provided in section 9.7 of the Appendix. The uncertainty parameter κ is a scalar that multiplies the variance described in Section 2. This parameter ranges from 0.5 to 2; 0.5 represents a farmer with a relative high degree of certainty, whereas 2 introduces more uncertainty to the farmer.

Estimating the coefficient of relative risk aversion in farmers has been a frequent topic of inquiry among agricultural economists, and the findings have identified a large amount

[1]The stochastic nature of the function, combined with its nonlinearity, cause the output to be quite sensitive to step tolerances and optimality tolerances. Robustness checks for these parameters are detailed in Sections 9.1-9.5 of the Appendix

of heterogeneity. Specifically, Chavas and Holt (1996) found that both corn and soybean farmers are significantly risk averse and exhibit decreasing absolute risk aversion. Moreover, Lence (2000) found that farmers' relative risk aversion has decreased throughout the 20th century; in particular, confidence intervals of risk aversion span from 1.08 to 3.67 in their estimation. Additionally, Friend and Blume (1975) estimated coefficients of relative risk aversion fall between 1 and 3. In light of these findings, we set our coefficients of relative risk aversion to fall between 1 and 3.

To relax the restrictive assumption of normally distributed priors, our representative farmer solves the optimization model with a combination of moment parameters that are amenable to the Fleishman power method. Berg (1988) proved that the cube of a normal distribution is indeterminate. Given the cubic nature of the Fleishman power method detailed in equation (5), this poses problems for certain combinations of moments. The parameters γ_1 and γ_2 are meticulously chosen to eschew this issue. More details of this selection process are provided in section 9.8 of the Appendix.

Table 1: Parameter Values Used in Optimization Model

Parameter	Values Considered
Risk Aversion (ρ)	$\{1,\ 2,\ 3\}$
Initial wealth (W_0)	$\{25,\ 50\}$
Optimism (o)	$\{0,\ 0.01\}$
Uncertainty (κ)	$\{0.5,\ 1,\ 1.5,\ 2\}$
Skewness (γ_1)	$\{0,\ 0.4,\ 0.8,\ 1\}$
Kurtosis (γ_2)	$\{-1,\ -0.5,\ 0,\ 0.5,\ 1\}$

Notes: Each parameter is varied individually while holding all remaining parameters fixed at their baseline values. The reported sets indicate the discrete values used to evaluate the sensitivity of optimal hedging decisions.

6 Results

Table 2 illustrates the results of the average final hedge ratios with respect to our structural parameters. We hold each parameter constant and allow the other parameters to fluctuate while assessing the mean of the season ending hedge ratio across 1000 iterations. Farmers who forecast just a 1 cent increase in the futures price each week choose not to hedge across the entirety of the risk aversion parameters, with the exception of $\rho = 3$, where an average of 3% of their crop is hedged. This shows how implementing optimism into our expected utility model can rationalize a farmer's decision not to hedge their corn in the futures market. Moreover, the level of hedging increases monotonically with risk aversion. This helps rationalizes the behavior of the small number of farmers who do hedge aggressively. Farmers with more uncertainty hedge more, as predicted by expected utility theory. Farmers with an uncertainty index of 0.5 hedge 83.9% of their crop, whereas farmers with an uncertainty level of 2 hedge nearly 8% more at 91.95%. Higher moments play an economically negligible role in hedging decisions. The season ending hedge ratio is 0.8121 at a skewness level of 0 and 0.8089 at a skewness level of 1. This is in accordance with our comparative statics results. Indeed, higher levels of skewness was predicted to decrease hedging; however, the effects are washed out by the Taylor coefficient and the ρ parameter in equation (17). For kurtosis, the season ending hedge ratio is 0.8805 in our most platykurtic distribution, and the season ending hedge ratio is 0.8852 in our most leptokurtic distribution. We expected to see an increase in hedging in the face of increasing kurtosis; however, similar to the skewness results, the Taylor coefficient and risk aversion parameter mute the effects quite significantly. Initial wealth effects shows that the effects of DARA hold in the absence of optimism; indeed, the optimal hedge ratio drops 3.3% as initial wealth increases from 25 to 50. All of these results are in accordance with our comparative statics in section 4. Mean and variance have noticeable changes to season ending hedge ratios, thus dominating the effects of higher order moments such as skewness and kurtosis.

Table 2: Season-Ending Hedge Ratios Across Model Specifications

Category	**Parameter**	**No Optimism**	**Optimism**
Risk Aversion (ρ)	1	0.8318	0.0032
	2	0.8838	0.0024
	3	0.9349	0.0362
Uncertainty (κ)	0.5	0.8390	0.0132
	1	0.8730	0.0118
	1.5	0.9009	0.0134
	2	0.9195	0.0193
Skewness (γ_1)	0	0.8121	0.0107
	0.4	0.8107	0.0109
	0.8	0.8095	0.0109
	1	0.8089	0.0109
Kurtosis (γ_2)	-1	0.8805	0.0138
	-0.5	0.8829	0.0142
	0	0.8841	0.0146
	0.5	0.8849	0.0147
	1	0.8852	0.0147
Wealth (W_0)	25	0.9002	0.0059
	50	0.8668	0.0229

Figures 5-6 show the effects of different parameter combinations as they relate to the final, optimal hedge ratio. Figure 5 shows that hedging increases monotonically with both risk aversion and uncertainty for risk averse farmers. Figure 6 illustrates the magnitude of the Decreasing Absolute Risk Aversion (DARA) effect, a well known characteristic of the CRRA utility function.

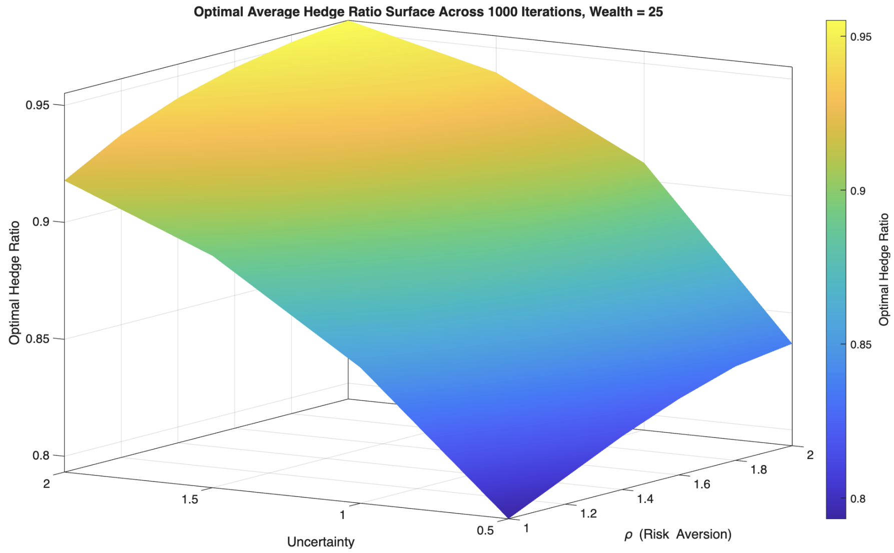

Figure 5: Risk Averse Farmers vs Uncertainty

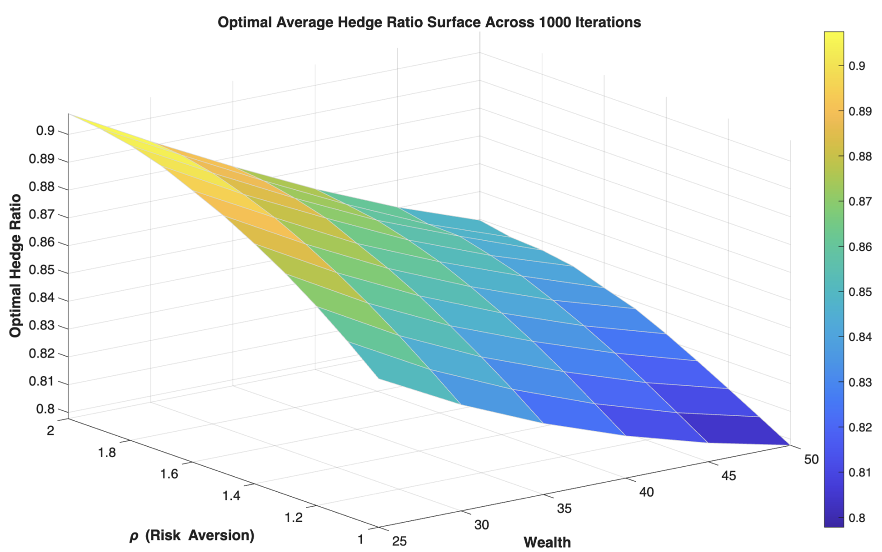

Figure 6: DARA Illustrated

7 Sequential Hedging

In the spirit of Anderson and Danthine (1983), we characterize the time path of our representative farmer's hedging decisions across different levels of risk aversion and uncertainty. We analyze how these utility function parameters influence when farmers decide to sell their corn in the futures market before the December contract expires. Over 1000 iterations, we assess how their season ending hedge fluctuates across different parameters. We split the season into cumulative thirds. Specifically, the first graph in each figure will show the distribution of hedge ratios from weeks 1-9, the second graph will show the distribution of hedge ratios from weeks 1-18, and the third graph will show the distribution of hedge ratios from weeks 1-27. In this section, we only analyze prior distributions that are normally distributed. When we endow our farmers with a tepid amount of optimism, they cease to hedge whatsoever, as illustrated in Table 2. Thus, we only characterize the time paths of farmers who decide to hedge and have an optimism (o) level of 0. The average hedge ratio of these graphs are mentioned below, and they are reported in further detail in section 9.6 of the Appendix.

7.1 $\rho = 1$ (Log Utility)

Under log utility, our representative farmer hedges the majority of their corn in the futures market, but tends to wait until the end of the season to do so. The log utility is not very concave compared to the functions we will look at below. Thus, Figures 7 and 8 show that the farmer with log utility tends to forego hedging until later in the season. In Figure 8, we see that increasing uncertainty causes the farmer to hedge more earlier in the season. This is consistent with expected utility behavior; indeed, higher uncertainty means more potential losses, and for the risk averse farmer, the magnitude of potential losses exceeds the magnitude of potential gains.

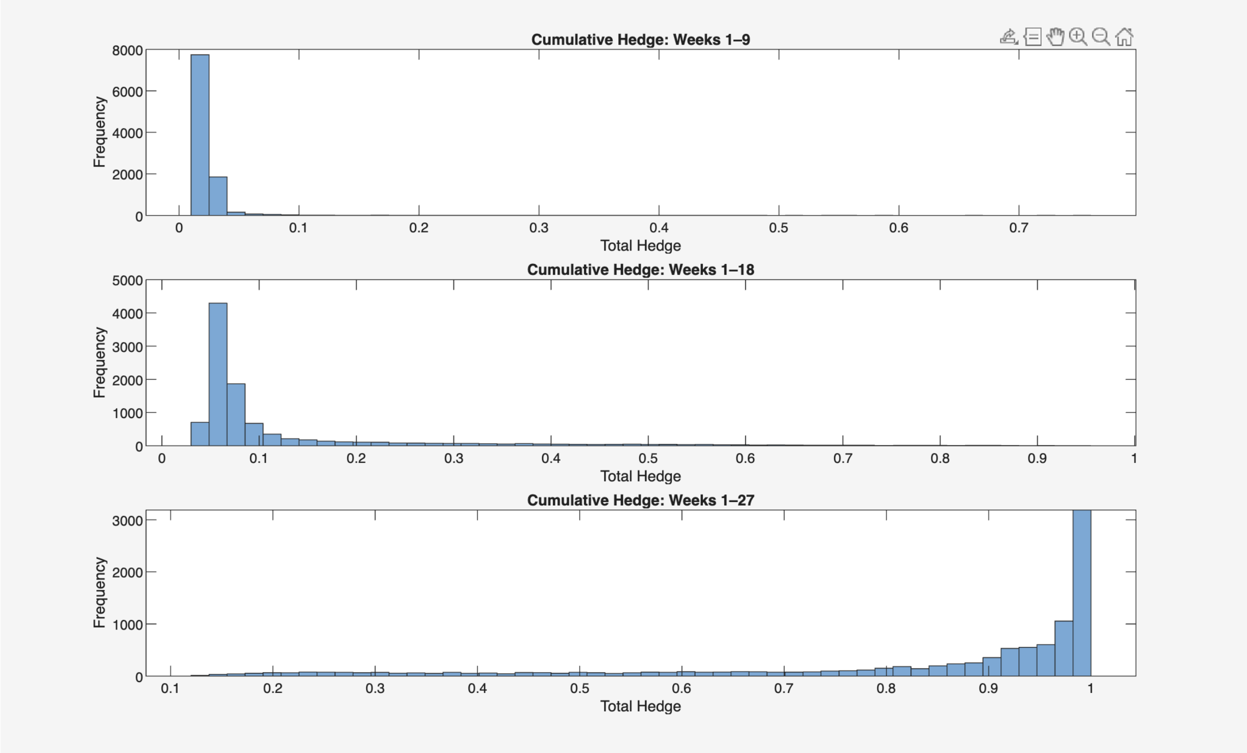

Figure 7: $\rho = 1$, Uncertainty $= 1$

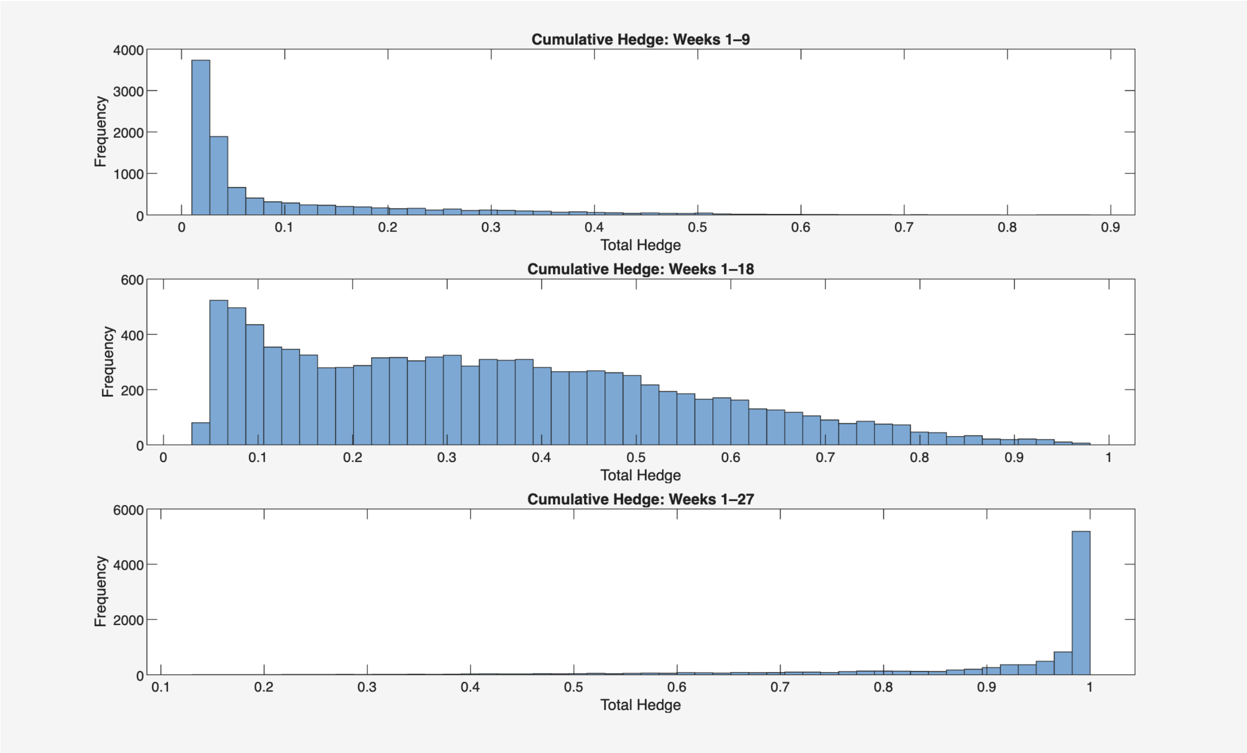

Figure 8: $\rho = 1$, Uncertainty $= 2$

7.2 $\rho = 2$ (Moderate Risk Aversion)

As we increase risk aversion, and thus the concavity of the utility function, the farmer hedges more earlier in the season. This is consistent with expected utility theory - early hedging helps lock in certain prices and helps alleviate uncertainty that is disliked by the risk averse farmer. When uncertainty = 1, the farmer hedges over 33% of their corn in the first 2/3 of the season. Under log utility, the farmer only hedged about 10% of their corn in the same time period. Moreover, increasing uncertainty leads to aggressive, early hedging when $\rho = 2$. Specifically, the farmer sells nearly half of their crop in the futures market in the first third of the season to hedge against potential downside exposure. Figures 9 and 10 illustrate the cumulative hedge path of our representative farmer when $\rho = 2$. Increasing the uncertainty induces the farmer to hedge more of their crop earlier on in the season.

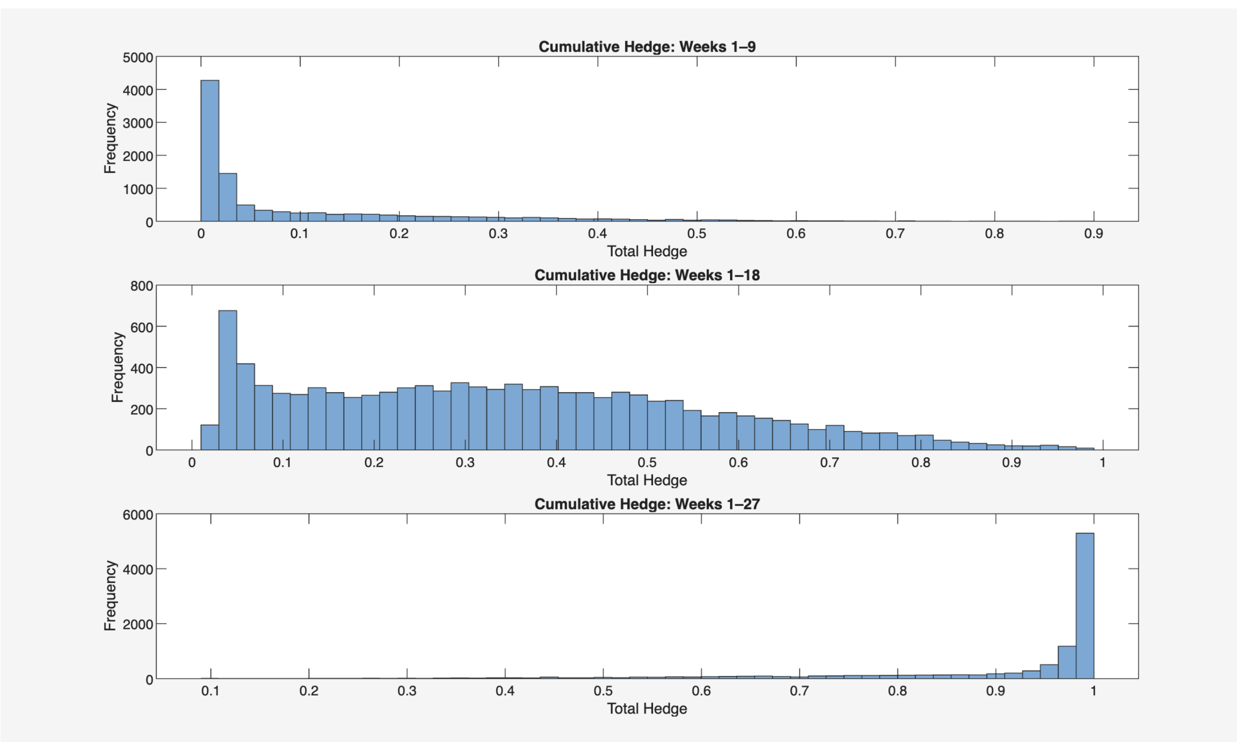

Figure 9: $\rho = 2$, Uncertainty = 1

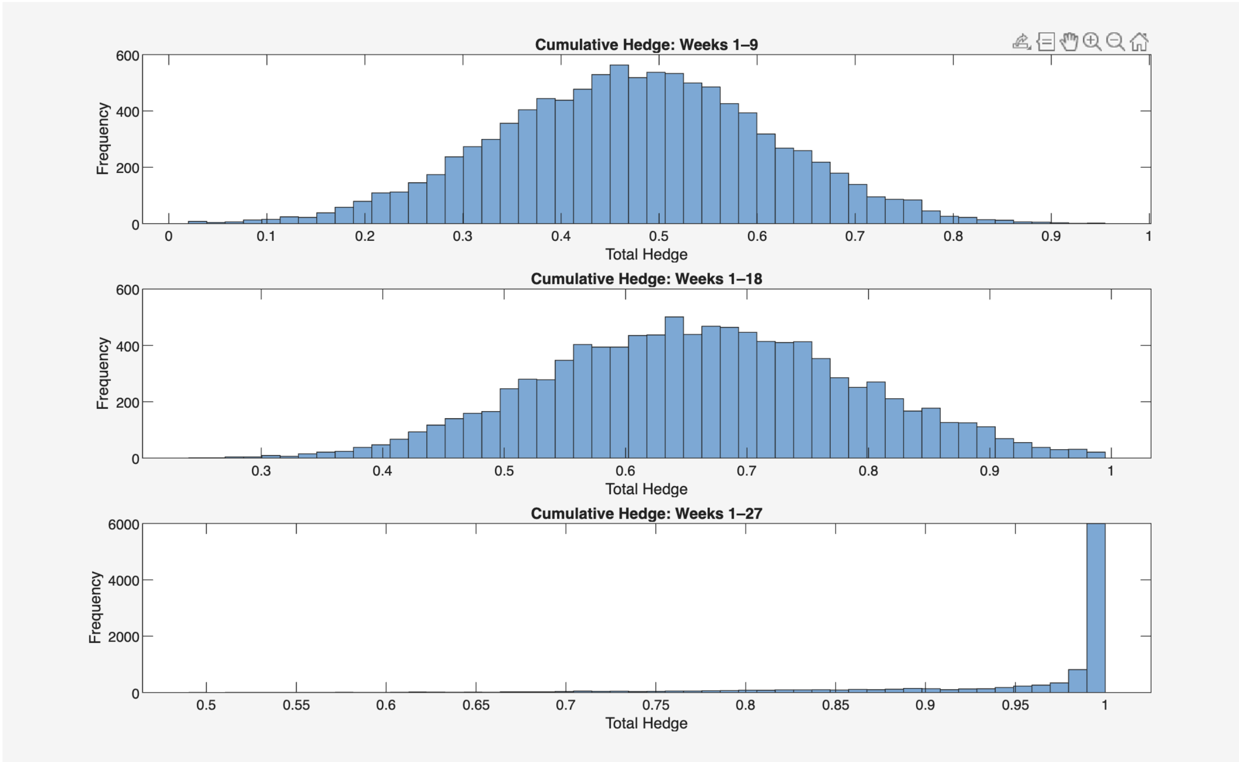

Figure 10: $\rho = 2$, Uncertainty $= 2$

7.3 $\rho = 3$, Strong Risk Aversion

Lastly, we analyze our most risk averse farmer. When $\rho = 3$, the total hedge at the end of the season is statistically negligible from the total the total hedge when $\rho = 2$. For both levels of risk aversion, the total hedging at the end of the season is approximately $94 - 95\%$, and the hedge increases to about 98% when uncertainty is increased. However, the time path of the hedge differs quite significantly. Specifically, when $\rho = 3$, the farmer hedges nearly 30% of their crop in the first third of the season. When $\rho = 2$, they only hedge 11% in the same time frame. Figures 11 and 12 illustrate the cumulative hedge path of our representative farmer when $\rho = 3$. Similar to when $\rho = 2$, the increased uncertainty induces the farmer to hedge more of their crop earlier on in the season.

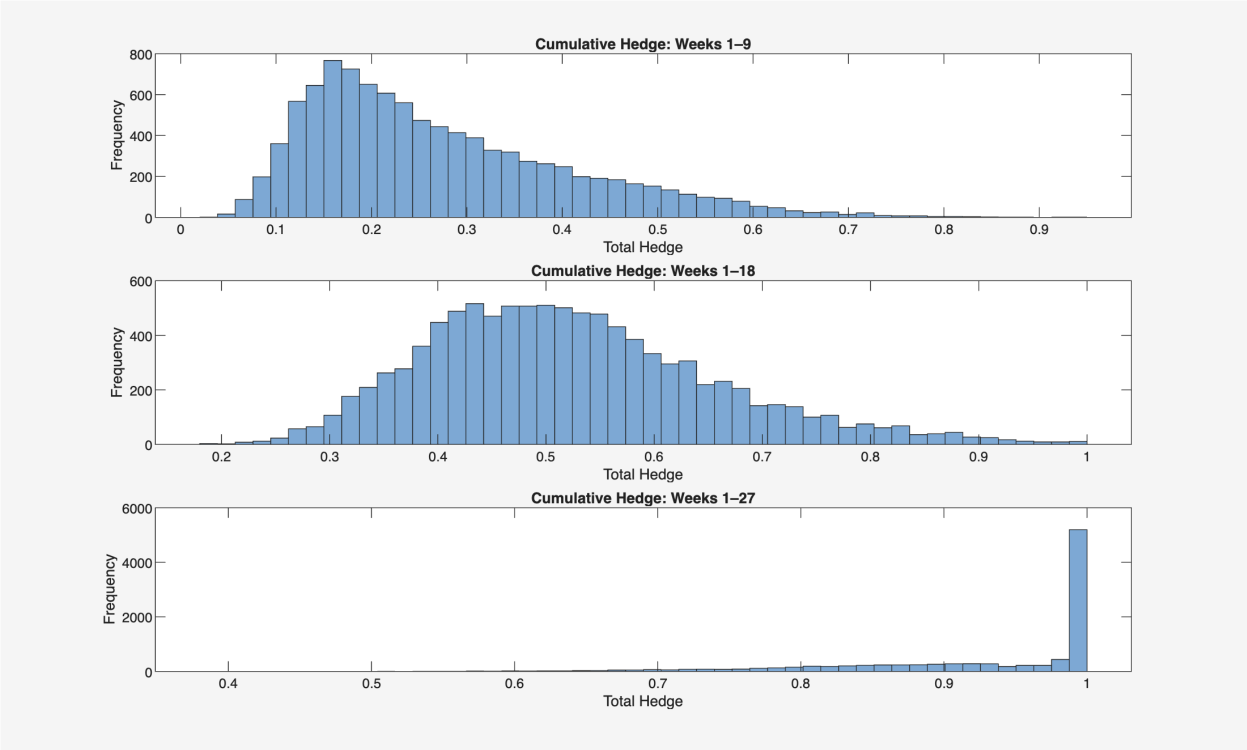

Figure 11: $\rho = 3$, Uncertainty $= 1$

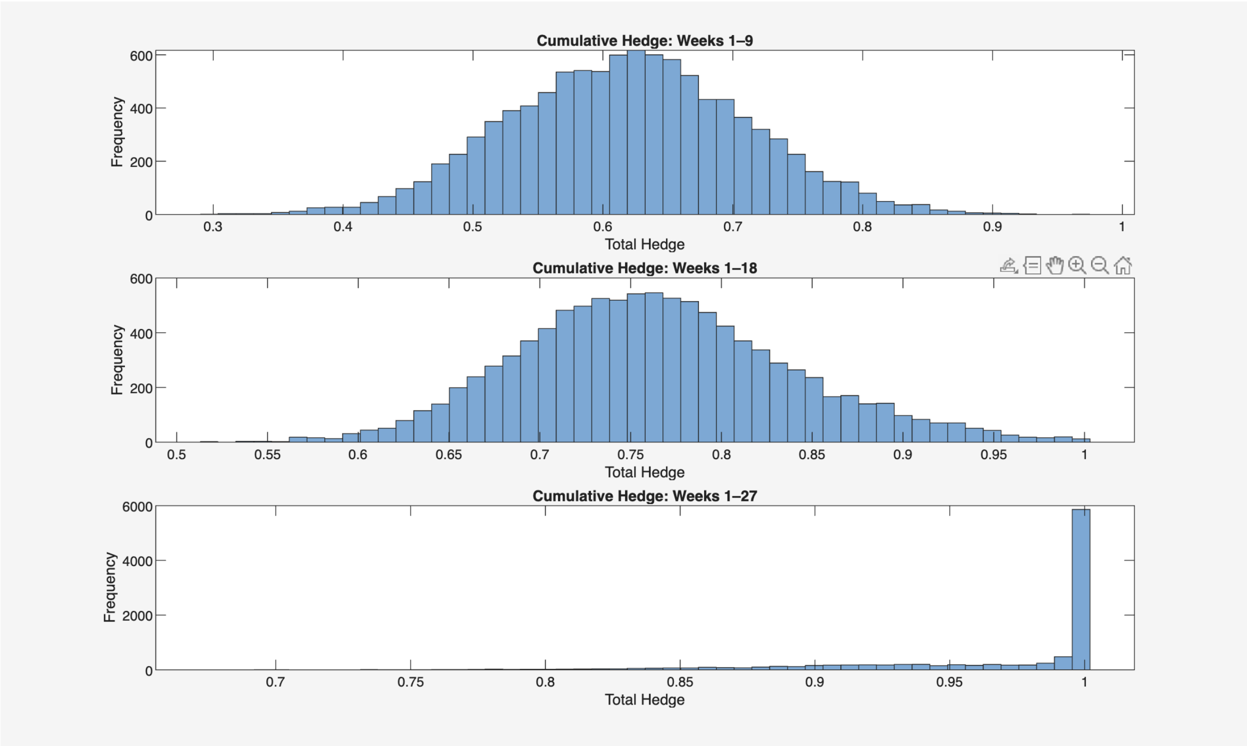

Figure 12: $\rho = 3$, Uncertainty $= 2$

8 Conclusion

Conventional mean-variance hedging theory can accurately rationalize the behavior of the farmers who aggressively hedge their corn in the futures market; however, the same framework fails to rationalize the behavior of the vast majority of farmers who forego hedging. When we endow our farmers with an optimism parameter in their utility function, they forego hedging completely, regardless of the risk aversion parameter. In other words, just a 1 cent weekly increase in the forecasted futures price results in virtually no hedging for all farmers across our array of risk aversion levels and other utility function parameters. These results remain robust to both Gaussian and non-Gaussian distributions. Indeed, many previous papers employ a CARA utility function with a normally distributed prior to exploit the closed form solution; however, our simulated results show that higher moments such as skewness and kurtosis have negligible impacts on season ending hedge ratios.

From the standpoint of sequential hedging, farmers with lower levels of risk aversion forego hedging until the end of the season, and higher prior variance encourages them to hedge more aggressively early on. Farmers with higher levels of risk aversion hedge more earlier in the season. Moreover, our analysis illustrates that farmers' hedging decisions are robust to different prior distribution assumptions. Our results illustrate that the seemingly irrational hedging patters of farmers can indeed be rationalized through expected utility analysis, as optimism about futures prices disincentives hedging and encourages farmers to sell their corn on the spot market.

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9 Appendix

9.1 Gradient Function

9.1.1 Gradient of the Objective Function

MATLAB's fmincon package is flexible towards specifying an objective gradient function. This improves both the efficiency of the algorithm and the accuracy of the optimization. The gradient function for each time period t is defined as such:

Revenue in period t is

$$R = W_0 + \sum_{t=1}^{T} f_t\theta_t + \sum_{k=1}^{t-1} f_k\theta_k + \sum_{j=t+1}^{T-1} (\tilde{f}_j + o)\tilde{\theta}_j + \left(1 - \sum_{t=1}^{T} \theta_t - \sum_{j=t+1}^{T-1} \tilde{\theta}_j\right) \tilde{P}_T.$$

Expected utility is

$$f(\theta,\tilde{\theta}) = \mathbb{E}\left[\frac{R^{1-\rho}}{1-\rho}\right].$$

The gradient used by the optimizer is

$$\frac{\partial f}{\partial \theta_t} = \mathbb{E}\Big[(f_t - \tilde{P}_T)R^{-\rho}\Big]$$

and for forward sales $j = t+1, \ldots, T-1$,

$$\frac{\partial f}{\partial \tilde{\theta}_j} = \mathbb{E}\Big[(\tilde{f}_j + o - \tilde{P}_T)R^{-\rho}\Big].$$

9.2 Optimization Parameters in MATLAB

The representative farmer's utility function becomes very flat at high levels of risk aversion. Thus, the default optimization parameters in MATLAB fail to provide meaningful results. We provide a sensitivity analysis of these parameters to find optimal values.

9.3 Optimality Tolerance

Let $\varepsilon_{\text{OptTol}}$ denote the optimality tolerance, $\mathscr{L}$ be the Lagrangian, and θ be the hedge ratio chosen by the farmer. If

$$\max_i \|\frac{\partial \mathscr{L}}{\partial \theta_i}\| \leq \epsilon_{OptTol} \tag{19}$$

Then the iterations stop. MATLAB's default value for optimality tolerance equals $1e^{-6}$

9.4 Step Tolerances

Let $\varepsilon_{\text{StepTol}}$ be the step tolerance. Step tolerance denotes the lower bound of the magnitude of a step on the x axis. Namely, $\|x_{t+1} - x_t\|_\infty$. In the fmincon package we employ, Step

Tolerance is relative bound. Thus, MATLAB halts the optimization algorithm when

$$\|x_{t+1} - x_t\|_\infty < \varepsilon_{\text{StepTol}} * (1 + |x_i|) \tag{20}$$

MATLAB's default value for step tolerances = $1e^{-10}$

9.5 Chosen Optimization Parameters

To assess optimal optimization parameters, we run 100 simulations across a grid of StepTol = $[10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}]$ and OptTol = $[10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}]$. The chosen optimization parameter is identified at the largest value of OptTol and StepTol such that the hedge ratio is stabilized.

9.5.1 $\rho = 1$

Table 3: Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 1$)

ρ	Optimism	OptTol	StepTol	Mean Hedge	Δ Mean Hedge	Notes
1	0	1.00×10^{-6}	1.00×10^{-10}	0.8209	–	Baseline
1	0	1.00×10^{-6}	1.00×10^{-6}	0.8209	0	
1	0	1.00×10^{-6}	1.00×10^{-8}	0.8209	0	
1	0	1.00×10^{-6}	1.00×10^{-10}	0.8209	0	
1	0	1.00×10^{-6}	1.00×10^{-12}	0.8209	0	
1	0	1.00×10^{-6}	1.00×10^{-10}	0.8209	0	
1	0	1.00×10^{-8}	1.00×10^{-10}	0.8148	8.32×10^{-6}	
1	0	1.00×10^{-10}	1.00×10^{-10}	0.8149	8.38×10^{-6}	
1	0	1.00×10^{-12}	1.00×10^{-10}	0.8149	8.38×10^{-6}	
1	0.01	1.00×10^{-6}	1.00×10^{-10}	0.0028	–	Baseline
1	0.01	1.00×10^{-6}	1.00×10^{-6}	0.0029	0	
1	0.01	1.00×10^{-6}	1.00×10^{-8}	0.0028	0	
1	0.01	1.00×10^{-6}	1.00×10^{-10}	0.0028	0	
1	0.01	1.00×10^{-6}	1.00×10^{-12}	0.0028	0	
1	0.01	1.00×10^{-6}	1.00×10^{-10}	0.0028	0	
1	0.01	1.00×10^{-8}	1.00×10^{-10}	2.83×10^{-5}	8.67×10^{-6}	
1	0.01	1.00×10^{-10}	1.00×10^{-10}	8.42×10^{-6}	8.70×10^{-6}	
1	0.01	1.00×10^{-12}	1.00×10^{-10}	8.30×10^{-6}	8.70×10^{-6}	

Notes: Table reports robustness of the optimal hedge ratio to alternative optimization tolerances. Baseline specification uses OptTol = 10^{-6} and StepTol = 10^{-10}. ΔMean Hedge reports the change relative to the baseline solution.

For $\rho = 1$, optimal hedge ratios are insensitive to changes in optimization parameters. Thus, the baseline is chosen.

9.5.2 $\rho = 2$

Table 4: Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 2$)

ρ	Optimism	OptTol	StepTol	Mean Hedge	Δ Mean Utility	Notes
2	0	1.00×10^{-6}	1.00×10^{-10}	0.9386	–	Baseline
2	0	1.00×10^{-6}	1.00×10^{-6}	0.9388	0	
2	0	1.00×10^{-6}	1.00×10^{-8}	0.9386	0	
2	0	1.00×10^{-6}	1.00×10^{-10}	0.9386	0	
2	0	1.00×10^{-6}	1.00×10^{-12}	0.9386	0	
2	0	1.00×10^{-6}	1.00×10^{-10}	0.9386	0	
2	0	1.00×10^{-8}	1.00×10^{-10}	0.8930	4.22×10^{-6}	
2	0	1.00×10^{-10}	1.00×10^{-10}	0.8993	4.26×10^{-6}	
2	0	1.00×10^{-12}	1.00×10^{-10}	0.9004	4.26×10^{-6}	
2	0.01	1.00×10^{-6}	1.00×10^{-10}	0.1896	–	Baseline
2	0.01	1.00×10^{-6}	1.00×10^{-6}	0.1896	0	
2	0.01	1.00×10^{-6}	1.00×10^{-8}	0.1896	0	
2	0.01	1.00×10^{-6}	1.00×10^{-10}	0.1896	0	
2	0.01	1.00×10^{-6}	1.00×10^{-12}	0.1896	0	
2	0.01	1.00×10^{-6}	1.00×10^{-10}	0.1896	0	
2	0.01	1.00×10^{-8}	1.00×10^{-10}	6.83×10^{-4}	8.88×10^{-6}	
2	0.01	1.00×10^{-10}	1.00×10^{-10}	1.15×10^{-5}	8.95×10^{-6}	
2	0.01	1.00×10^{-12}	1.00×10^{-10}	8.56×10^{-6}	8.95×10^{-6}	

Notes: Table reports robustness of the optimal hedge ratio to alternative optimization tolerances. Baseline specification uses OptTol $= 10^{-6}$ and StepTol $= 10^{-10}$. ΔMean Hedge reports the change relative to the baseline solution.

At $\rho = 2$, the hedge ratio decreases by 4.56% at OptTol $= 10^{-8}$, then stabilizes. Similarly, at StepTol $= 10^{-8}$ with positive optimism, the optimal hedge ratio decreases nearly 19%. Because both of these changes yield small but positive expected utility increases, we choose OptTol $= 10^{-8}$ and StepTol $= 10^{-8}$.

9.5.3 $\rho = 3$

Table 5: Tolerance Robustness Check for Mean Hedge Ratio ($\rho = 3$)

ρ	Optimism	OptTol	StepTol	Mean Hedge	Δ Mean Utility
3	0	1.00×10^{-6}	1.00×10^{-10}	0.9625	–
3	0	1.00×10^{-6}	1.00×10^{-6}	0.9073	-3.12×10^{-9}
3	0	1.00×10^{-6}	1.00×10^{-8}	0.9625	0
3	0	1.00×10^{-6}	1.00×10^{-10}	0.9625	0
3	0	1.00×10^{-6}	1.00×10^{-12}	0.9625	0
3	0	1.00×10^{-6}	1.00×10^{-10}	0.9625	0
3	0	1.00×10^{-8}	1.00×10^{-10}	0.9169	2.30×10^{-7}
3	0	1.00×10^{-10}	1.00×10^{-10}	0.9114	2.60×10^{-7}
3	0	1.00×10^{-12}	1.00×10^{-10}	0.9116	2.60×10^{-7}
3	0.01	1.00×10^{-6}	1.00×10^{-10}	0.8899	–
3	0.01	1.00×10^{-6}	1.00×10^{-6}	0.8555	-1.75×10^{-8}
3	0.01	1.00×10^{-6}	1.00×10^{-8}	0.8899	-5.61×10^{-11}
3	0.01	1.00×10^{-6}	1.00×10^{-10}	0.8899	0
3	0.01	1.00×10^{-6}	1.00×10^{-12}	0.8810	-2.01×10^{-8}
3	0.01	1.00×10^{-6}	1.00×10^{-10}	0.8899	0
3	0.01	1.00×10^{-8}	1.00×10^{-10}	0.0499	3.19×10^{-6}
3	0.01	1.00×10^{-10}	1.00×10^{-10}	0.0031	3.26×10^{-6}
3	0.01	1.00×10^{-12}	1.00×10^{-10}	0.0026	3.26×10^{-6}

Notes: Table reports robustness of the optimal hedge ratio to alternative optimization tolerances. Baseline specification uses OptTol $= 10^{-6}$ and StepTol $= 10^{-10}$. ΔMean Hedge reports the change relative to the baseline solution.

At $\rho = 3$, the utility function is quite concave, thus warranting a much smaller step size. We see that, under positive optimism, results stability around 0.0031 when OptTol and StepTol both equal 10^{-10}. Thus, both of these parameters are chosen.

9.6 Optimal Hedge Ratios by Time Period

9.6.1 $\rho = 1$

Table 6: Average Hedging by Season Third ($\rho = 1$, Wealth = 25, Uncertainty = 1)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.0057	0.0362
Mid Season (10–18)	0.0943	0.1898
Late Season (19–27)	0.8221	0.2565
Total Hedge	0.9221	

Table 7: Average Hedging by Season Third ($\rho = 1$, Wealth = 25, Uncertainty = 2)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.1130	0.1634
Mid Season (10–18)	0.2584	0.2492
Late Season (19–27)	0.5897	0.2673
Total Hedge	0.9611	

9.6.2 $\rho = 2$

Table 8: Average Hedging by Season Third ($\rho = 2$, Wealth = 25, Uncertainty = 1)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.1130	0.1477
Mid Season (10–18)	0.2519	0.2286
Late Season (19–27)	0.5805	0.2434
Total Hedge	0.9454	

Table 9: Average Hedging by Season Third ($\rho = 2$, Wealth = 25, Uncertainty = 2)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.4919	0.1365
Mid Season (10–18)	0.1795	0.1564
Late Season (19–27)	0.3047	0.1351
Total Hedge	0.9760	

9.6.3 $\rho = 3$

Table 10: Average Hedging by Season Third ($\rho = 3$, Wealth = 25, Uncertainty = 1)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.2939	0.1437
Mid Season (10–18)	0.2511	0.1509
Late Season (19–27)	0.3965	0.1452
Total Hedge	0.9415	

Table 11: Average Hedging by Season Third ($\rho = 3$, Wealth = 25, Uncertainty = 2)

Season Period (t)	**Mean Total Hedge**	**Standard Deviation**
Early Season (1–9)	0.6302	0.0893
Mid Season (10–18)	0.1383	0.0866
Late Season (19–27)	0.2098	0.0817
Total Hedge	0.9783	

9.7 Initial Wealth Calculation

We assume that farmers' initial wealth is a function of land value and investments in farming equipment. In 2024, corn farmers yielded about 179.3 bushels per acre (USDA, NASS (2025)). Additionally, the average corn farm contained about 466 acres on average (Lacy (2025)). Moreover, land value was found to be approximately \$4,170 per acre (Callahan (2026)). Machinery investment value was found to be \$847 / acre for farms that were less than 500 acres (Langemeier (2025)). Thus, total land value can be appoximated as 466 * \$4,170 = \$1,943,220. Total equipment value can be calculated as \$847 * 466 = \$394,702. After combining land value and equipment value, our representative farmer has a total wealth of \$1,943,220 + \$394,702 = \$2,337,922. Thus, wealth per bushel can be approximated as \$2,337,922 / 83,553.80 ≈ \$27.98. Thus, we simplify and use \$25 as a baseline wealth level. Clearly, farmers often have income from outside of the farm as well. Thus, we use \$50 as the initial wealth level for farmers with substantial off-farm income.

9.8 Fleishman Power Method Robustness Test

Table 19 illustrates the Fleishman Power method across our grid of higher moments when both mean and variance are fixed at 0 and 1, respectively. Specifically, column 1 shows the intended skewness parameter, and column 2 shows the intended kurtosis parameter. Columns 3, 4, 5, 6 show all four moments resulting from the generated Fleishman distribution.

Table 12: Moment Matching Diagnostics for Fleishman Power Method

γ_1	γ_2	Mean	Variance	Skewness	Excess Kurtosis
0	-1	0.009	1.001	−0.016	−0.999
0	-0.5	0.014	1.005	−0.016	−0.468
0	0	−0.015	1.023	−0.005	−0.028
0	0.5	0.008	0.973	0.004	0.516
0	1	−0.011	0.990	0.013	0.859
0.4	-1	0.011	0.982	0.384	−0.959
0.4	-0.5	0.006	0.982	0.388	−0.496
0.4	0	0.009	0.979	0.372	−0.079
0.4	0.5	−0.006	0.989	0.333	0.490
0.4	1	0.009	1.019	0.424	0.740
0.8	-1	0.010	0.865	0.787	−0.230
0.8	-0.5	0.007	0.942	0.791	−0.196
0.8	0	0.000	1.005	0.803	−0.022
0.8	0.5	0.009	0.990	0.741	0.316
0.8	1	−0.001	0.972	0.763	0.891
1	-1	−0.011	0.767	0.985	0.439
1	-0.5	−0.020	0.854	1.060	0.901
1	0	−0.007	0.927	0.977	0.368
1	0.5	−0.008	0.991	0.999	0.460
1	1	−0.011	0.957	0.994	1.202

9.9 Sensitivity Analysis

Table 13: Sensitivity of Optimal Hedging Decisions to Model Parameters

Parameter	Average Sensitivity Measure
Optimism (Mean Shift)	0.7995
Risk Aversion (ρ)	0.0997
Wealth	0.0291
Uncertainty (Variance)	0.0283
Skewness	0.0019
Kurtosis	0.0019

Notes: Each sensitivity measure is computed by varying one parameter at a time while holding all other parameters fixed, and then averaging the resulting change in the optimal hedge position. Larger values indicate greater influence on hedging behavior.